

MANSI PATIL

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, CGPA 8.74/10) with project experience across data analytics, machine learning, and Generative AI. Built ML models for prediction and forecasting (Scikit-learn, XGBoost), Retrieval-Augmented Generation (RAG) pipelines with vector search (Qdrant), and LLM-powered applications (Groq, OpenAI, Claude APIs) using prompt engineering and embeddings. Comfortable with SQL, Python, PySpark, Power BI, and data visualization (Streamlit/Plotly dashboards), and tracked model experiments in MLflow for reproducibility. Seeking an entry-level Data & Analytics Engineer role to build semantic models, dashboards, data pipelines, and AI-enabled solutions.

TECHNICAL SKILLS

BI, Analytics & Data: Power BI, SQL, Python, PySpark, Pandas, NumPy, Feature Engineering, Data Cleansing & Validation, Matplotlib/Seaborn
Generative AI / LLM: LLM Integration, Prompt Engineering, RAG, Multimodal RAG, Embeddings, Vector Databases (Qdrant), Cross-Modal Retrieval, LangChain, OpenAI GPT, Claude, Gemini, Groq API, Hugging Face
Machine Learning: Scikit-learn, XGBoost, PyTorch, Model Evaluation & Cross-Validation, OpenCV, YOLO
Backend, APIs & Databases: FastAPI, Django, REST APIs, MySQL, PostgreSQL, SQLite
Tools & Workflow Automation: Docker, Git/GitHub, GitHub Actions, MLflow (experiment tracking), Jupyter, Streamlit, Postman

EXPERIENCE

Data Science Intern | Amdox Technologies

Apr 2026 – Jun 2026

NeuralRetail — AI Sales Intelligence Platform | [GitHub](#)

- **Engineered** an enterprise retail analytics system integrating 3 ML models — XGBoost churn prediction (0.84+ ROC-AUC), ARIMA 30-day demand forecasting, and K-Means customer segmentation.
- **Architected** secure RESTful FastAPI endpoints with JWT authentication and SQLite persistence, powering real-time model scoring and automated inventory reorder alerts.
- **Built** an interactive Streamlit/Plotly dashboard visualizing sales trends and churn simulations, and tracked lifecycle metrics for all 3 models with MLflow, supporting reproducible experimentation and model monitoring.

PROJECTS (DATA, ML & GENERATIVE AI FOCUS)

Multimodal RAG Engine — Cross-Modal Document Intelligence System | [GitHub](#)

Python, FastAPI, Qdrant, CLIP, BLIP-2, Claude API, Docker

- **Built** a multimodal RAG pipeline that extracts text, tables, and high-DPI page images from PDFs using PyMuPDF, with sentence-aware chunking (~300 chars) to eliminate mid-sentence breaks.
- **Designed** a dual-vector retrieval system combining SentenceTransformer (384-d) text embeddings and OpenCLIP (512-d) image embeddings via Reciprocal Rank Fusion, enabling joint retrieval of text and visual chart content.
- **Reduced** context noise by applying a dynamic relevance cutoff (≥ 0.40) and MS-MARCO Cross-Encoder reranking in Qdrant, cutting retrieved context to 3 high-precision chunks per query.
- **Engineered** a grounded Vision-LLM generator (Claude API) with XML-based prompt-injection defenses, inline page citations, and refusal logic to prevent hallucinated answers.
- **Deployed** a FastAPI backend with threadpool-based background execution and startup model pre-warming, paired with a Streamlit UI and an automated evaluation suite for regression testing.

Newsly — AI-Powered Daily News Digest Agent | [GitHub](#) | [Live Demo](#)

Python, FastAPI, Groq API (Llama 3.3-70B), Docker, GitHub Actions

- **Created** an LLM-powered news agent retrieving and summarizing articles across 5 categories in real time using Google News RSS for retrieval and prompt-engineered summarization with Llama 3.3-70B — an applied RAG-style pipeline.
- **Automated** a FastAPI dashboard with scheduled Gmail SMTP digest delivery; containerized with Docker and orchestrated daily runs via GitHub Actions.

Loan Approval Prediction System with AI Chatbot | [GitHub](#) | [Live Demo](#)

Python, Scikit-learn, XGBoost, FastAPI, Streamlit, OpenAI API (GPT-4o-mini)

- **Delivered** an end-to-end ML classification pipeline (XGBoost, Scikit-learn) to assess applicant financial risk and predict loan approval status based on 10+ demographic and credit variables, applying supervised learning and model evaluation best practices.
- **Integrated** an AI financial advisory chatbot powered by the OpenAI API (GPT-4o-mini) to explain prediction rationale and provide actionable, data-grounded recommendations.
- **Deployed** an interactive Streamlit analytics dashboard alongside the full-stack system, taking it live to cloud production via Render.

EDUCATION

Maharashtra Institute of Technology — B.Tech, Computer Science Engineering

CGPA: 8.74 / 10

CERTIFICATIONS

- Business Intelligence with Advanced Excel and Power BI (2024)
- CSRBOX — Agentic AI: From Learner to Builder – AI Agent Architect (2025)
- Deloitte — Data Analyst Certificate (2025)